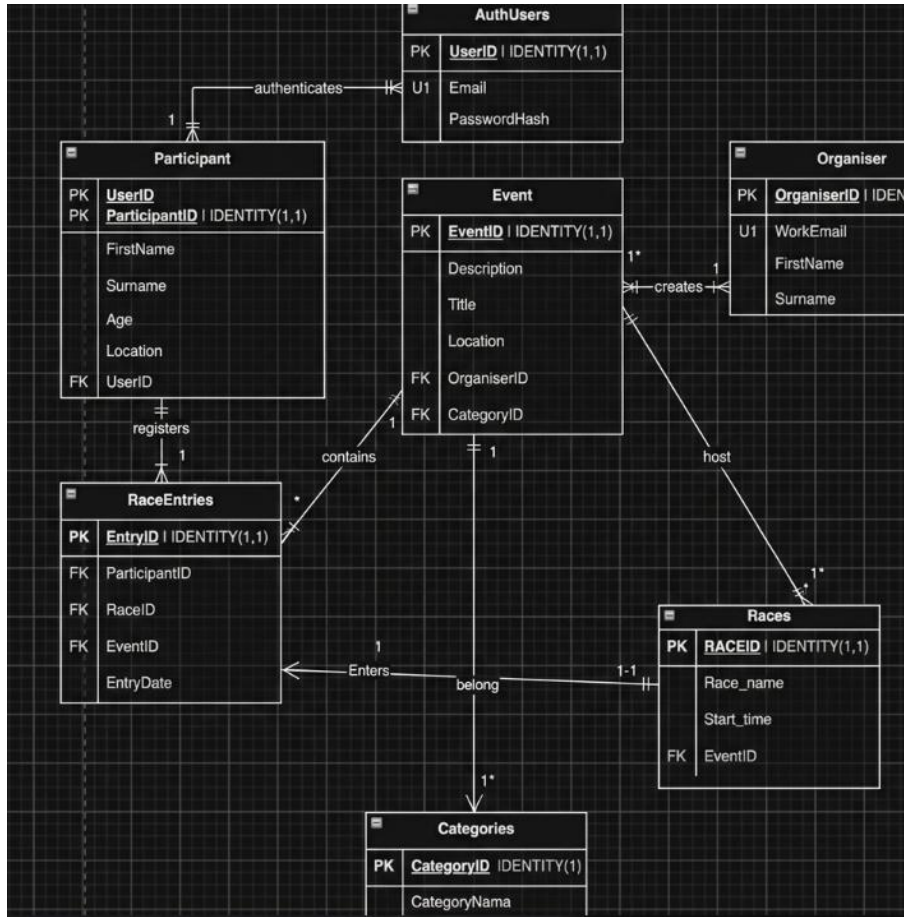


Here is our business rules that will guide our ERD

- An organiser can create and manage multiple events.
- Each organiser's name and surname must be stored in the database.
- An organiser can create multiple race events.
- participant must register their profile using password,username,email before being allowed to enter event.
- A participant may enter only one race at a time, as races within an event may take place simultaneously.
- Each participant's name, surname, age, and location must be stored in the database.
- Each event can contain multiple races.
- Each event must include a title, description, and city must be stored in the database.
- Each Event must belong to one category, such as cycling, walking and running.
- Each event takes place in a specific city

ERD Diagram

2



Step 1 Create table representing by ERD

ERD Update

We add this line because we had to update the ERD to include authenticated table because assignment needs us to make sure participants register to login

3

```
1  USE master;
2  GO
3
4  IF EXISTS (SELECT * FROM sys.databases WHERE name = 'RacedayDb')
5  BEGIN
6      ALTER DATABASE RacedayDb SET SINGLE_USER WITH ROLLBACK IMMEDIATE;
7      DROP DATABASE RacedayDb;
8  END
9  GO
10
11 CREATE DATABASE RacedayDb;
12 GO
13 USE RacedayDb;
14 GO
15 |
```

Tables

```

GO
USE RacedayDb;
GO

-- Drop child tables first to avoid foreign key dependency errors
DROP TABLE IF EXISTS RaceEntries;
DROP TABLE IF EXISTS Races;
DROP TABLE IF EXISTS Event;
DROP TABLE IF EXISTS Organiser;
DROP TABLE IF EXISTS Categories;
DROP TABLE IF EXISTS Participant;
DROP TABLE IF EXISTS AuthUsers;
GO

CREATE TABLE AuthUsers (
    UserID INT IDENTITY(1,1) NOT NULL,
    Username VARCHAR(50) NOT NULL,
    PasswordHash VARCHAR(255) NOT NULL,
    Email VARCHAR(100) NOT NULL,
    CONSTRAINT PK_AuthUsers PRIMARY KEY (UserID)
);

CREATE TABLE Participant (
    ParticipantID INT IDENTITY(1,1) NOT NULL,
    FirstName VARCHAR(50) NOT NULL,
    Surname VARCHAR(50) NOT NULL,
    Age INT NOT NULL,
    Location VARCHAR(100) NOT NULL,
    UserID INT NULL,
    CONSTRAINT PK_Participant PRIMARY KEY (ParticipantID),
    CONSTRAINT FK_Participant_AuthUsers FOREIGN KEY (UserID) REFERENCES AuthUsers(UserID)
);

CREATE TABLE Categories (
    CategoryID INT IDENTITY(1,1) NOT NULL,
    CategoryName VARCHAR(50) NOT NULL,
    CONSTRAINT PK_Categories PRIMARY KEY (CategoryID)
);

CREATE TABLE Organiser (
    OrganiserID INT IDENTITY(1,1) NOT NULL,
    FirstName VARCHAR(50) NOT NULL,
    Surname VARCHAR(50) NOT NULL,
    CONSTRAINT PK_Organiser PRIMARY KEY (OrganiserID)
);

```

5

```
CREATE TABLE Event (  
    EventID INT IDENTITY(1,1) NOT NULL,  
    Description VARCHAR(255) NOT NULL,  
    Title VARCHAR(100) NOT NULL,  
    Location VARCHAR(100) NOT NULL,  
    OrganiserID INT NOT NULL,  
    CategoryID INT NOT NULL,  
    CONSTRAINT PK_Event PRIMARY KEY (EventID),  
    CONSTRAINT FK_Event_Organiser FOREIGN KEY (OrganiserID) REFERENCES Organiser(OrganiserID),  
    CONSTRAINT FK_Event_Categories FOREIGN KEY (CategoryID) REFERENCES Categories(CategoryID)  
);  
  
CREATE TABLE Races (  
    RaceID INT IDENTITY(1,1) NOT NULL,  
    RaceName VARCHAR(50) NOT NULL,  
    StartTime DATETIME NOT NULL,  
    EventID INT NOT NULL,  
    CONSTRAINT PK_Races PRIMARY KEY (RaceID),  
    CONSTRAINT FK_Races_Event FOREIGN KEY (EventID) REFERENCES Event(EventID)  
);  
  
CREATE TABLE RaceEntries (  
    EntryID INT IDENTITY(1,1) NOT NULL,  
    ParticipantID INT NOT NULL,  
    RaceID INT NOT NULL,  
    EventID INT NOT NULL,  
    EntryDate DATETIME NOT NULL,  
    CONSTRAINT PK_RaceEntries PRIMARY KEY (EntryID),  
    CONSTRAINT FK_RaceEntries_Participant FOREIGN KEY (ParticipantID) REFERENCES Participant(ParticipantID),  
    CONSTRAINT FK_RaceEntries_Races FOREIGN KEY (RaceID) REFERENCES Races(RaceID),  
    CONSTRAINT FK_RaceEntries_Event FOREIGN KEY (EventID) REFERENCES Event(EventID)  
);  
GO
```

Step 2: Define keys and constraints

List of Entities and their attributes

1. AuthUsers (UserID PRIMARY KEY,

Username VARCHAR(50) NOT NULL,

PasswordHash VARCHAR (255) NOT NULL,

Email VARCHAR (100) NOT NULL,

CONSTRAINT PK_AuthUsers PRIMARY KEY (UserID));

2. Organiser

OrganiserID PRIMARY KEY,

FirstName VARCHAR (20) NOT NULL,

Surname VARCHAR (20) NOT NULL,

3. Participant

participantID PRIMARY KEY,

```

FirstName VARCHAR (20) NOT NULL,
Surname VARCHAR 20) NOT NULL,
Age int NOT NULL,
Location VARCHAR (20) NOT NULL,

```

4. Categories

```

CategoryID PRIMARY KEY,
CategoryName VARCHAR(50) NOT NULL,

```

5. Events

- [Event]

```

EventID INT IDENTITY(1,1) PRIMARY KEY,
[Description] VARCHAR(255) NOT NULL,
Title VARCHAR(90) NOT NULL,
[Location] VARCHAR(20) NULL,
OrganiserID INT NOT NULL,
CategoryID INT NOT NULL,
CONSTRAINT FK_Event_Organiser FOREIGN KEY (OrganiserID) REFERENCES
Organiser(OrganiserID),
CONSTRAINT FK_Event_Category FOREIGN KEY (CategoryID) REFERENCES
Categories(CategoryID)
);

```

Commented [NM1]: Alter table to make title come first

6. Races

```

RaceID INT IDENTITY(1,1) PRIMARY KEY,
RaceName VARCHAR(50) NOT NULL,
StartTime DATETIME NOT NULL,
EventID INT NOT NULL,

```

7. RaceEntries

```

EntryID INT IDENTITY(1,1) PRIMARY KEY,
ParticipantID INT NOT NULL,
RaceID INT NOT NULL,
EventID INT NOT NULL,
EntryDate DATETIME NOT NULL DEFAULT GETDATE(),
CONSTRAINT FK_RaceEntries_Participant FOREIGN KEY (ParticipantID)
REFERENCES Participant(ParticipantID),
CONSTRAINT FK_RaceEntries_Race FOREIGN KEY (RaceID) REFERENCES
Races(RaceID),
CONSTRAINT FK_RaceEntries_Event FOREIGN KEY (EventID) REFERENCES
[Event](EventID) ON DELETE CASCADE,

-- Enforces: "A participant may enter only one race per event at a
time "
CONSTRAINT UQ_Participant_Single_Race_Per_Event UNIQUE
(ParticipantID, EventID)

```

Step 3 : Seed Data

7

INSERT sample records

```
-- 5. Re-seed Clean Dataset (Safe from Duplicate Key Errors)

SET IDENTITY_INSERT AuthUsers ON;
INSERT INTO AuthUsers (UserID, Username, PasswordHash, Email)
SELECT 1, 'Markrosch', 'TK1855812', 'MK18@gmail.com' WHERE NOT EXISTS (SELECT 1 FROM AuthUsers WHERE UserID = 1);
INSERT INTO AuthUsers (UserID, Username, PasswordHash, Email)
SELECT 2, 'Pasily', 'TK185581200', 'pasily@gmail.com' WHERE NOT EXISTS (SELECT 1 FROM AuthUsers WHERE UserID = 2);
SET IDENTITY_INSERT AuthUsers OFF;

SET IDENTITY_INSERT Participant ON;
INSERT INTO Participant (ParticipantID, FirstName, Surname, Age, Location, UserID)
SELECT 1, 'Naked1', 'Montsho', 24, 'Johannesburg', 1 WHERE NOT EXISTS (SELECT 1 FROM Participant WHERE ParticipantID = 1);
INSERT INTO Participant (ParticipantID, FirstName, Surname, Age, Location, UserID)
SELECT 3, 'Jene', 'Smith', 28, 'Pretoria', 2 WHERE NOT EXISTS (SELECT 1 FROM Participant WHERE ParticipantID = 3);
INSERT INTO Participant (ParticipantID, FirstName, Surname, Age, Location, UserID)
SELECT 4, 'John', 'Doe', 29, 'Durban', NULL WHERE NOT EXISTS (SELECT 1 FROM Participant WHERE ParticipantID = 4);
INSERT INTO Participant (ParticipantID, FirstName, Surname, Age, Location, UserID)
SELECT 5, 'Sarah', 'Khnum', 31, 'Cape Town', NULL WHERE NOT EXISTS (SELECT 1 FROM Participant WHERE ParticipantID = 5);
SET IDENTITY_INSERT Participant OFF;

SET IDENTITY_INSERT Categories ON;
INSERT INTO Categories (CategoryID, CategoryName) SELECT 1, '15km Race' WHERE NOT EXISTS (SELECT 1 FROM Categories WHERE CategoryID = 1);
INSERT INTO Categories (CategoryID, CategoryName) SELECT 2, '5km Race' WHERE NOT EXISTS (SELECT 1 FROM Categories WHERE CategoryID = 2);
INSERT INTO Categories (CategoryID, CategoryName) SELECT 3, '5km Run' WHERE NOT EXISTS (SELECT 1 FROM Categories WHERE CategoryID = 3);
SET IDENTITY_INSERT Categories OFF;

SET IDENTITY_INSERT Organiser ON;
INSERT INTO Organiser (OrganiserID, FirstName, Surname) SELECT 1, 'David', 'Van Wyk' WHERE NOT EXISTS (SELECT 1 FROM Organiser WHERE OrganiserID = 1);
INSERT INTO Organiser (OrganiserID, FirstName, Surname) SELECT 2, 'Dane', 'Vasco' WHERE NOT EXISTS (SELECT 1 FROM Organiser WHERE OrganiserID = 2);
INSERT INTO Organiser (OrganiserID, FirstName, Surname) SELECT 3, 'John', 'Doe' WHERE NOT EXISTS (SELECT 1 FROM Organiser WHERE OrganiserID = 3);
INSERT INTO Organiser (OrganiserID, FirstName, Surname) SELECT 4, 'John', 'Doe' WHERE NOT EXISTS (SELECT 1 FROM Organiser WHERE OrganiserID = 4);
SET IDENTITY_INSERT Organiser OFF;

SET IDENTITY_INSERT Event ON;
INSERT INTO Event (EventID, Description, Title, Location, OrganiserID, CategoryID)
SELECT 1, 'Annual 10km city marathon through downtown', 'City Marathon 2026', 'Johannesburg', 1, 1 WHERE NOT EXISTS (SELECT 1 FROM Event WHERE EventID = 1);
INSERT INTO Event (EventID, Description, Title, Location, OrganiserID, CategoryID)
SELECT 2, 'Annual 15km city marathon through rural', 'Rural Marathon 2026', 'Magaliesburg', 1, 1 WHERE NOT EXISTS (SELECT 1 FROM Event WHERE EventID = 2);
INSERT INTO Event (EventID, Description, Title, Location, OrganiserID, CategoryID)
SELECT 3, 'Annual morning run', 'Summer Dash', 'Stadium A', 1, 1 WHERE NOT EXISTS (SELECT 1 FROM Event WHERE EventID = 3);
INSERT INTO Event (EventID, Description, Title, Location, OrganiserID, CategoryID)
SELECT 4, 'Annual cold run', 'Winter Dash', 'Stadium B', 1, 1 WHERE NOT EXISTS (SELECT 1 FROM Event WHERE EventID = 4);
SET IDENTITY_INSERT Event OFF;

2 SET IDENTITY_INSERT Races ON;
3 INSERT INTO Races (RaceID, RaceName, StartTime, EventID)
4 SELECT 1, '15km Race', '2008-05-05 00:00:00.000', 1 WHERE NOT EXISTS (SELECT 1 FROM Races WHERE RaceID = 1);
5 INSERT INTO Races (RaceID, RaceName, StartTime, EventID)
6 SELECT 2, '5km Race', '2008-05-05 00:00:00.000', 2 WHERE NOT EXISTS (SELECT 1 FROM Races WHERE RaceID = 2);
7 INSERT INTO Races (RaceID, RaceName, StartTime, EventID)
8 SELECT 3, 'Elite Wave', '2026-10-10 09:00:00.000', 1 WHERE NOT EXISTS (SELECT 1 FROM Races WHERE RaceID = 3);
9 INSERT INTO Races (RaceID, RaceName, StartTime, EventID)
0 SELECT 4, 'Open Wave...', '2026-10-10 09:00:00.000', 2 WHERE NOT EXISTS (SELECT 1 FROM Races WHERE RaceID = 4);
1 SET IDENTITY_INSERT Races OFF;
2
3 SET IDENTITY_INSERT RaceEntries ON;
4 INSERT INTO RaceEntries (EntryID, ParticipantID, RaceID, EventID, EntryDate)
5 SELECT 1, 1, 1, 1, '2008-05-05 07:00:00.000' WHERE NOT EXISTS (SELECT 1 FROM RaceEntries WHERE EntryID = 1);
6 SET IDENTITY_INSERT RaceEntries OFF;
7 GO
8
```

Step 4 testing script and Output

SQLQuery10.s...Student (64) Finalupdated...Student (70))* AllQueries....tudent (79))*

```

1  USE RacedayDb
2
3  SELECT * FROM AuthUsers;
4  SELECT * FROM Participant;
5  SELECT * FROM Categories;
6  SELECT * FROM Organiser;
7  SELECT * FROM Event;
8  SELECT * FROM RaceEntries;
9  SELECT * FROM Races;
10

```

74 % No issues found

Results Messages

UserID	Username	PasswordHash	Email
1	Markrosh	MK1855@12	MK1@gmail.com
2	Pasly	TK1855@1200	pasly@gmail.com

ParticipantID	FirstName	Surname	Age	Location	UserID
1	Nakedi	Montsho	24	Johannesburg	1
2	Jane	Smith	28	Pretoria	2
3	John	Doe	29	Durban	NULL
4	Sarah	Khnum	31	Cape Town	NULL

CategoryID	CategoryName
1	15km Race
2	5km Race
3	5km Run

OrganiserID	FirstName	Surname
1	David	Van Wyk
2	Dane	Vasco
3	John	Doe
4	John	Doe

EventID	Description	Title	Location	OrganiserID	CategoryID
1	Annual 10km city marathon through downtown.	City Marathon 2026	Johannesburg	1	1
2	Annual 15km city marathon through rural.	Rural Marathon 2026	Magaliesburg	1	1
3	Annual morning run	Summer Dash	Stadium A	1	1
4	Annual cold run	Winter Dash	Stadium B	1	1

EntryID	ParticipantID	RaceID	EventID	EntryDate
1	1	1	1	2008-05-05 07:00:00.000

RaceID	RaceName	StartTime	EventID
1	15km Race	2008-05-05 00:00:00.000	1
2	5km Race	2008-05-05 00:00:00.000	2
3	Elite Wave	2026-10-10 09:00:00.000	1
4	Open Wa...	2026-10-10 09:00:00.000	2